

PHASE STATISTICS APPROACH TO FINANCIAL TIME SERIES

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In this paper, an approach we introduced recently to study financial time series [1,2,3] is reviewed. The approach mainly consists of an application of the empirical mode decomposition to decompose an empirical time series into a number of intrinsic mode functions (IMFs), calculation of the instantaneous phase of the resultant IMFs, and the statistics of the instantaneous phase for each IMF. To illustrate the approach, we consider phase distribution and phase correlation of stock time series [1,3] and foreign exchange time series [2,3] as examples. The formulation of the approach is systematic and can be applied to the analysis of other time series.

Keywords

Phase statistics, empirical mode decomposition, time series analysis

References

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